



Faculty of Technology

New Generation Engineering...

BLM 2010

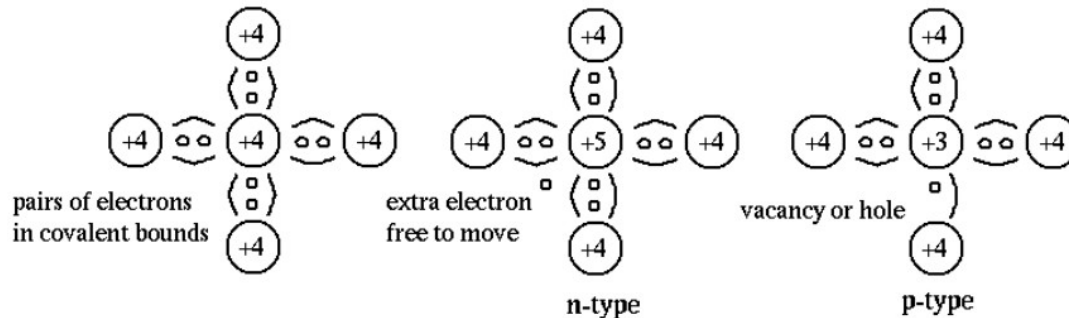
Elektronik Devrelere Giriş

Diyotlar

Dr. Öğr. Üyesi Timur İnan

p type-n type

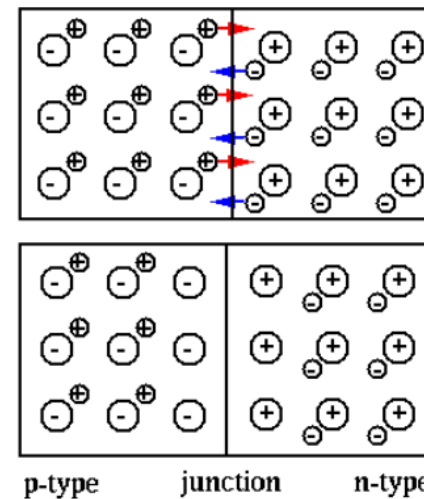
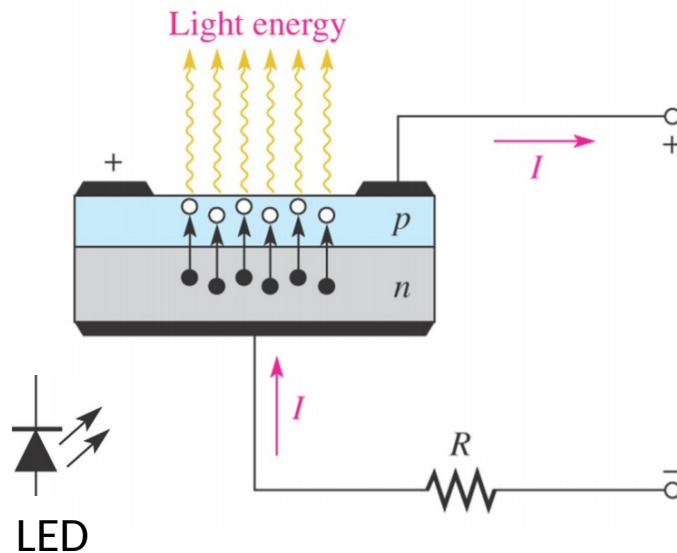
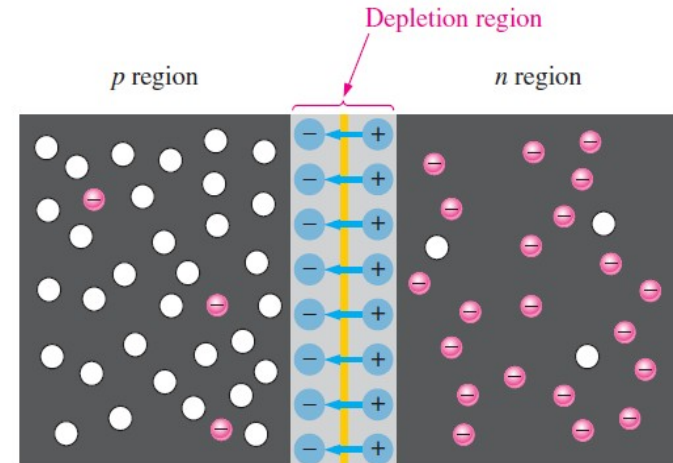
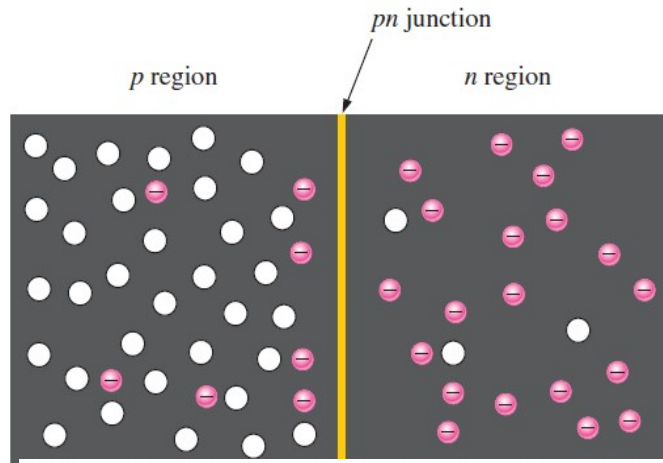
- **p tipi:** Son yörüngesinde **4** valans elektronu bulunan bir yarıiletkenin kristal örgüsünün içerisine Bor (B), Galyum (Ga), İndiyum (In),... gibi son yörüngesinde **3** valans elektrona sahip olan yabancı elementler eklenerek oluşturulur.
- **n tipi:** Son yörüngesinde **4** valans elektronu bulunan bir yarıiletkenin kristal örgüsünün içerisine Fosfor (P), Arsenik (As), Antimon (Sb),... gibi son yörüngesinde **5** valans elektrona sahip olan yabancı elementler eklenerek oluşturulur.



<http://fourier.eng.hmc.edu/e84/lectures/ch4/node1.html>

Y.T.Ü. Fen Edebiyat Fakültesi, Fizik Bölümü, Atom ve Moleküler Fiziği Laboratuvar Föyü

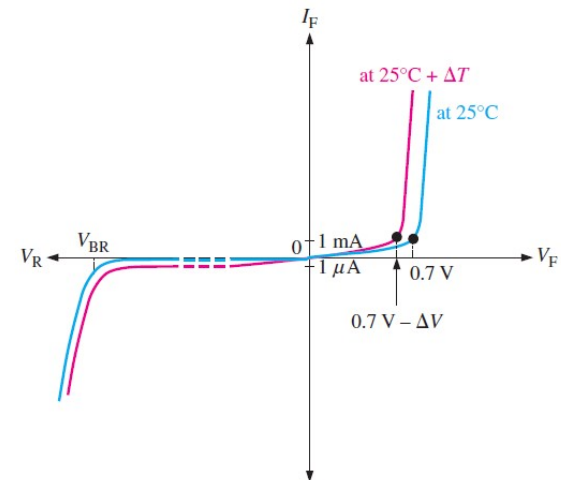
pn junction

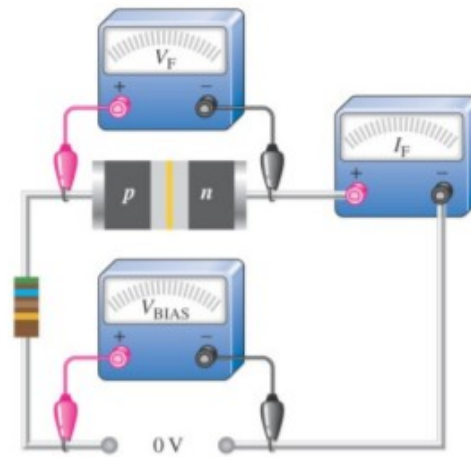


Diodes

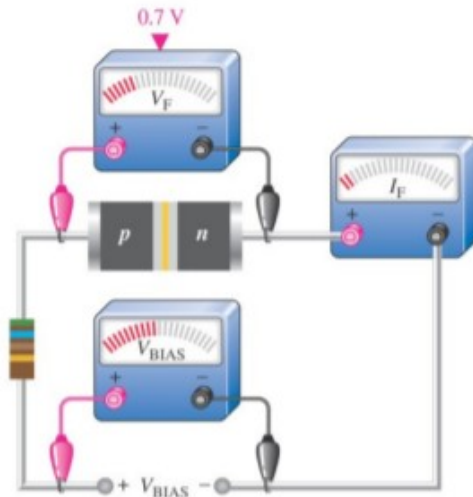
The PN Junction Diode

- The **barrier potential**, V_B , is the amount of voltage required to move electrons through the electric field
 - At 25°C, it is approximately 0.7 V for silicon and 0.3 V for germanium
 - As the junction temperature increases, the barrier potential decreases, and vice versa

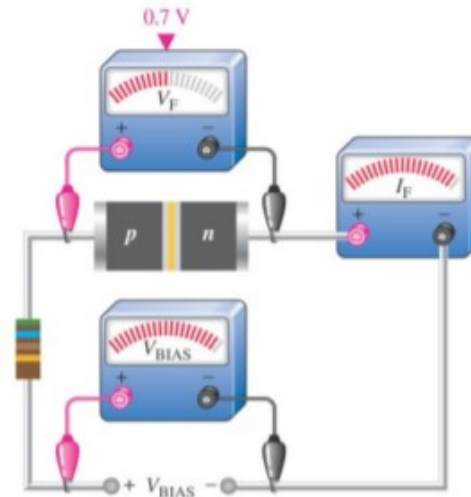




(a) No bias voltage. The pn junction of the diode is at equilibrium.

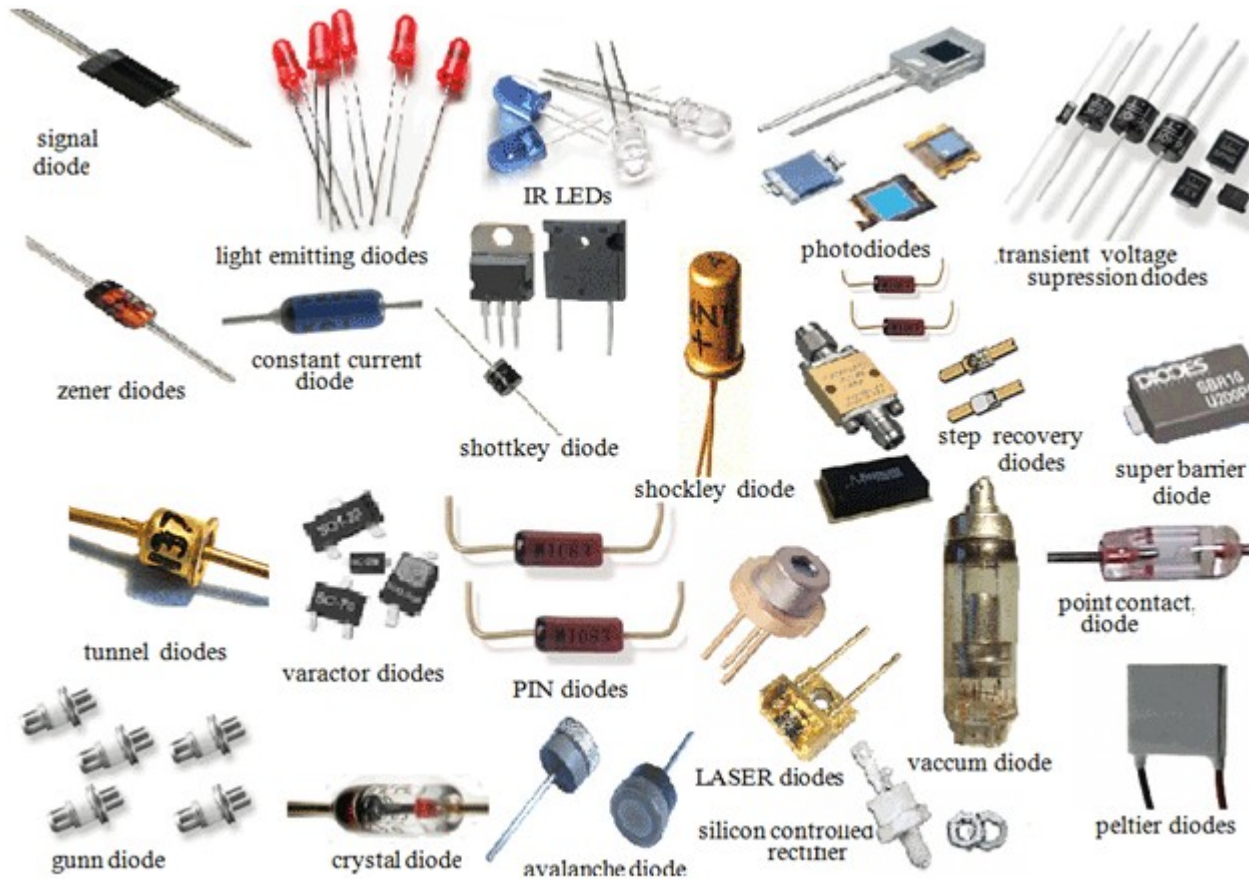


(b) Small forward-bias voltage ($V_F < 0.7$ V), very small forward current.



(c) Forward voltage reaches and remains at approximately 0.7 V. Forward current continues to increase as the bias voltage is increased.

Diodes



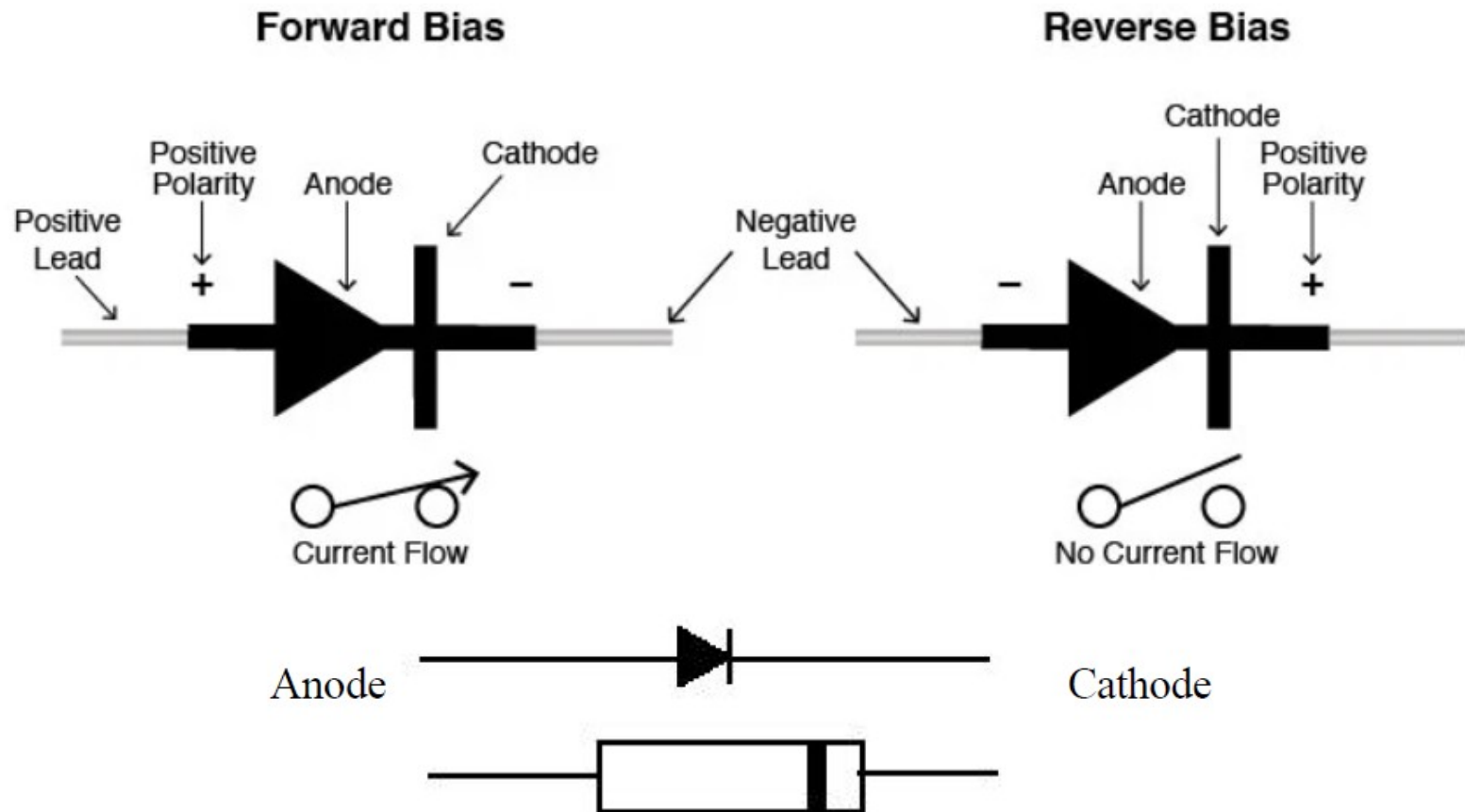
Types of Diode

Diodes



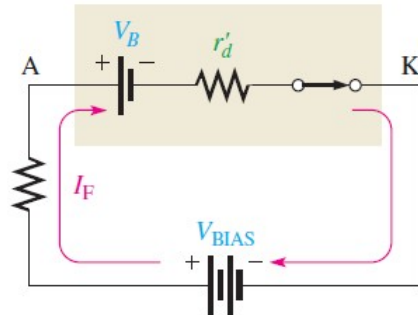
Diyotlar çeşitli konfigürasyonlarda mevcuttur. Soldan itibaren: metal kutu, çivi montajı, bantlı plastik kutu, kanallı plastik kutu, cam kutu.

Diodes

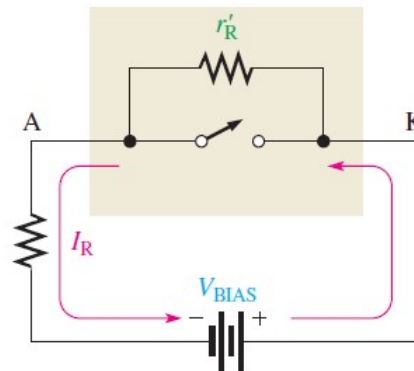


<https://www.fluke.com/tr-tr/bilgi-edinin/en-iyi-uygulamalar/olcum-temel-ilkeleri/elektrik/diyot-nedir>

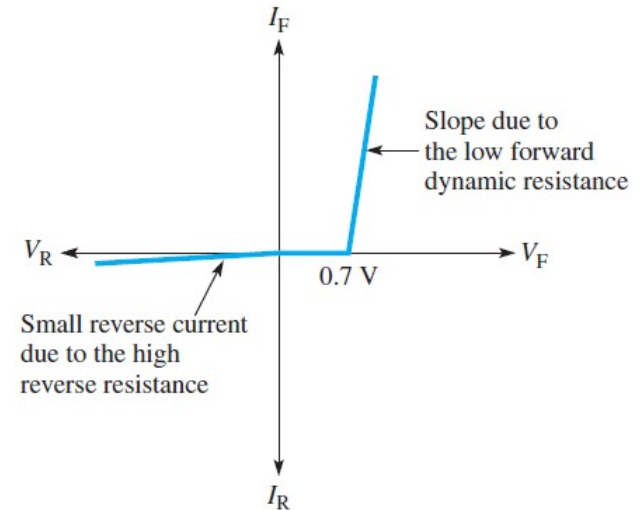
Real Forward-Bias / Reverse-Bias



(a) Forward bias



(b) Reverse bias

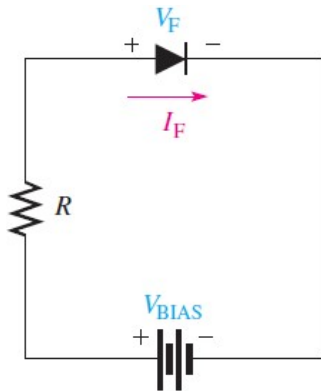


(c) V - I characteristic curve

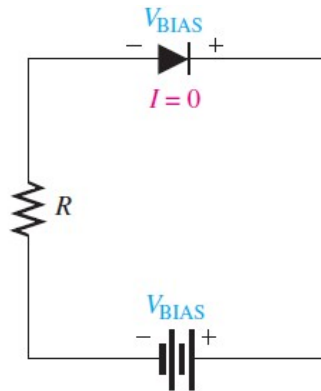
$$V_F = 0.7 \text{ V} + I_F r'_d$$

$$I_F = \frac{V_{\text{BIAS}} - 0.7 \text{ V}}{R_{\text{LIMIT}} + r'_d}$$

Forward-Bias vs Reverse-Bias



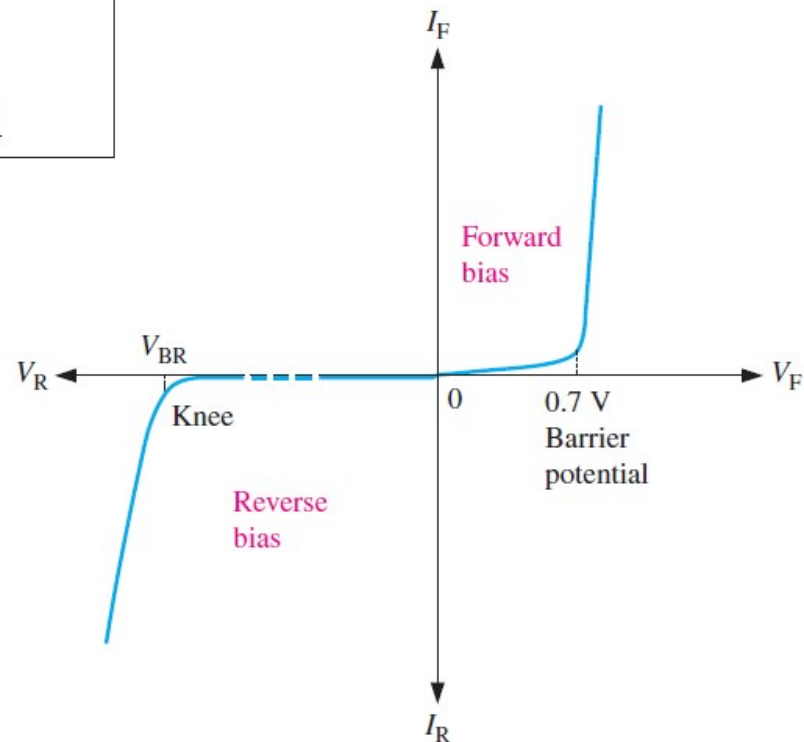
(a) Forward bias

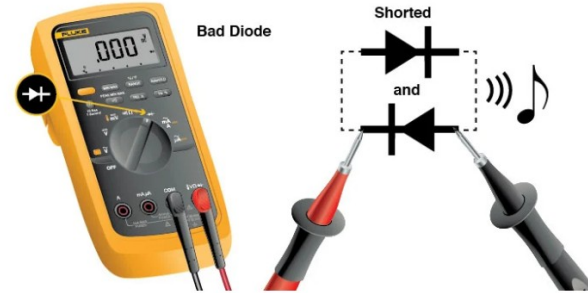
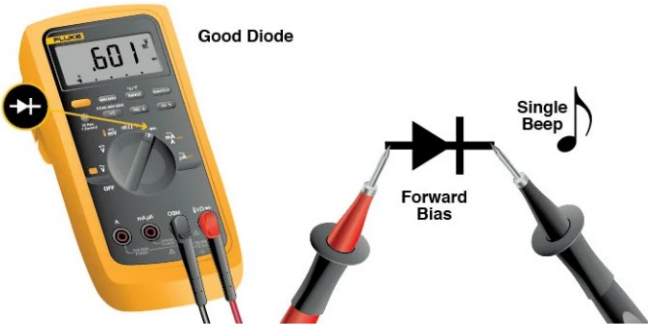
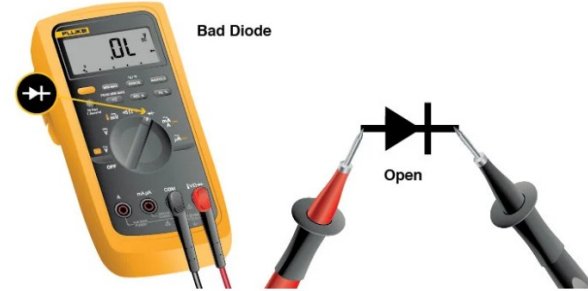
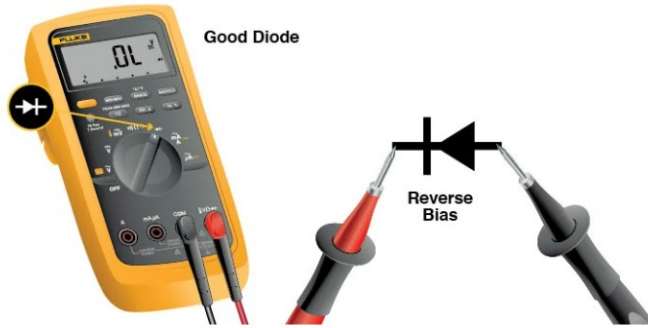


(b) Reverse bias

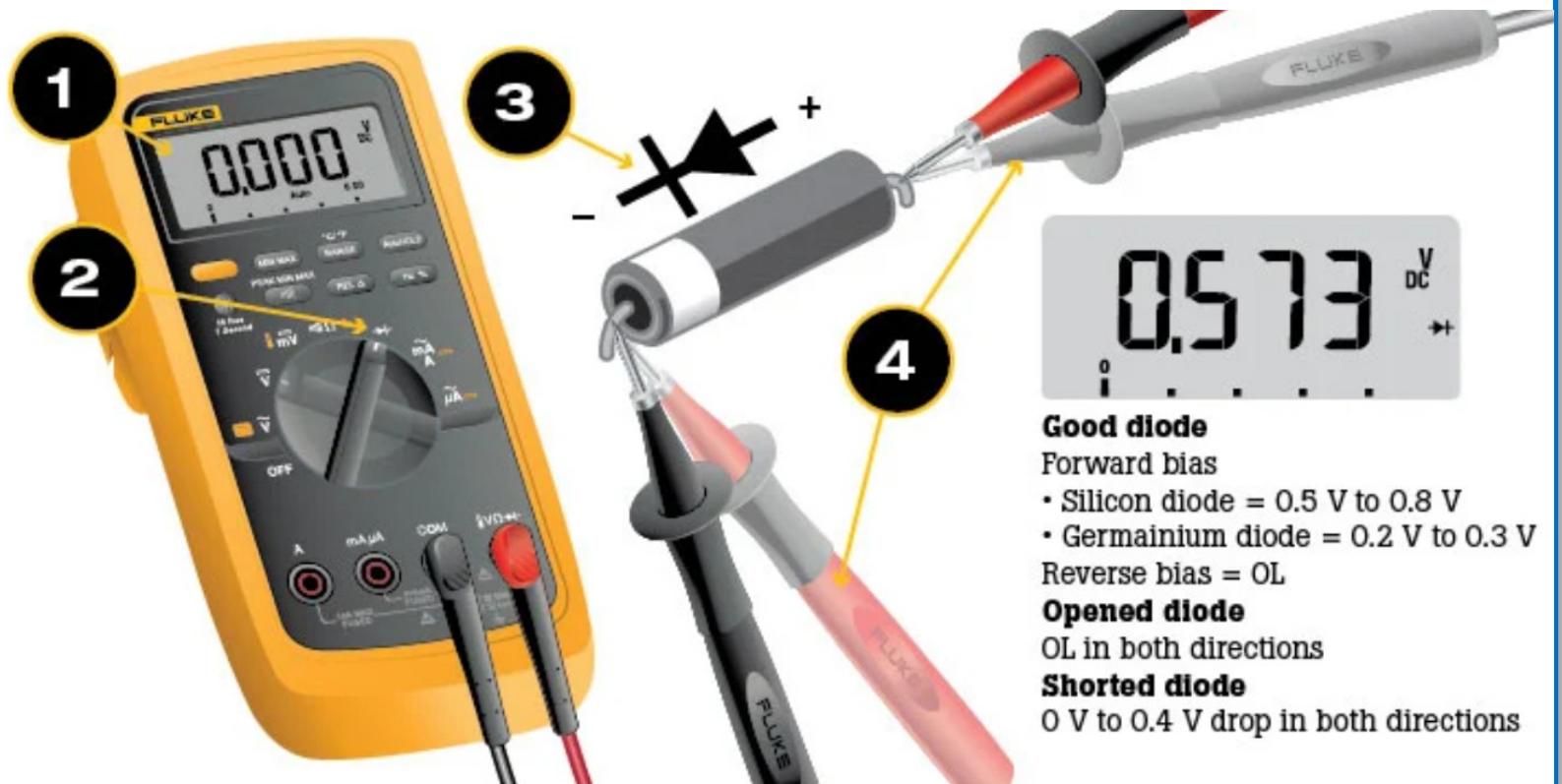
$$V_F = 0.7 \text{ V} + I_F r'_d$$

$$I_F = \frac{V_{\text{BIAS}} - 0.7 \text{ V}}{R_{\text{LIMIT}} + r'_d}$$





İyi durumda bir diyotun düz bias direnci 1000 ohm ile 10 ohm arasında değişmelidir. Ters bias olduğunda, Multimetrenin ekranında OL gösterilir (bu, çok yüksek direnç olduğunu gösterir).



Diyotun sađlamlık testi 2 řekilde yapılır:

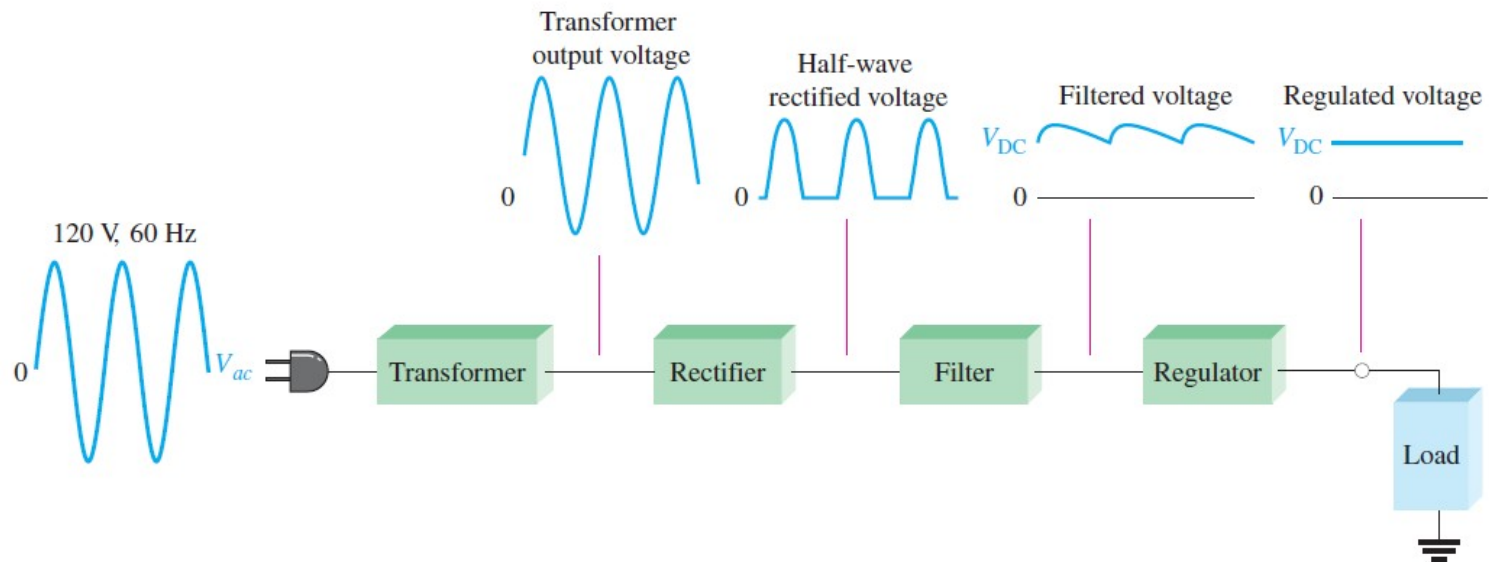
- *Ohmmetre ile; diyot bir yönde küçük direnç(300Ω - 3000Ω) ,problar ters takıldığında ise büyük direnç ($50k\Omega$ - $200k\Omega$) gösteriyorsa sağlamdır.*
- *Polarlama gerilimi ile; digital ölçü aletinin diyot kademesinde yapılan ölçümde bir yönde diyot üzerinde 0.2v-0.95v görülür, diğer yönde herhangi bir değer ölçülmez ise diyot sağlamdır.*
- *Diyot uçlarının belirlenmesi için öncelikle fiziki kontrol yapılır. Diyotun bir ucunda gri bant var ise bu katot ucudur.*
- *Gri bant silinmiş veya görülemiyor ise ohmmetre kullanılarak ölçüm yapılarak belirlenebilir. Diyotun bacakları belirlenirken ölçü aleti direnç ölçme kademesine getirilir ve diyotun uçları řu řekilde ölçülür: Düşük direnç ölçülen durumda ölçü aletinin siyah probun takılı olduğu uç katot (-) ucunu, kırmızı probun takılı olduğu uç diyotun anot (+) ucunu gösterir.*

Diyot Kullanırken Dikkat Edilmesi Gerekenler:

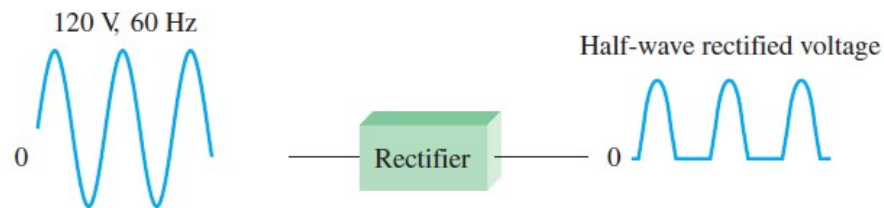
1. *Ters dayanma geriliminin üzerine çıkılmamalıdır.*
2. *Maksimum taşıma akımından daha fazla akım çekilmemelidir.*

http://www.ktu.edu.tr/dosyalar/trabzonelektrik_a78c0.pdf

DC Power Supply

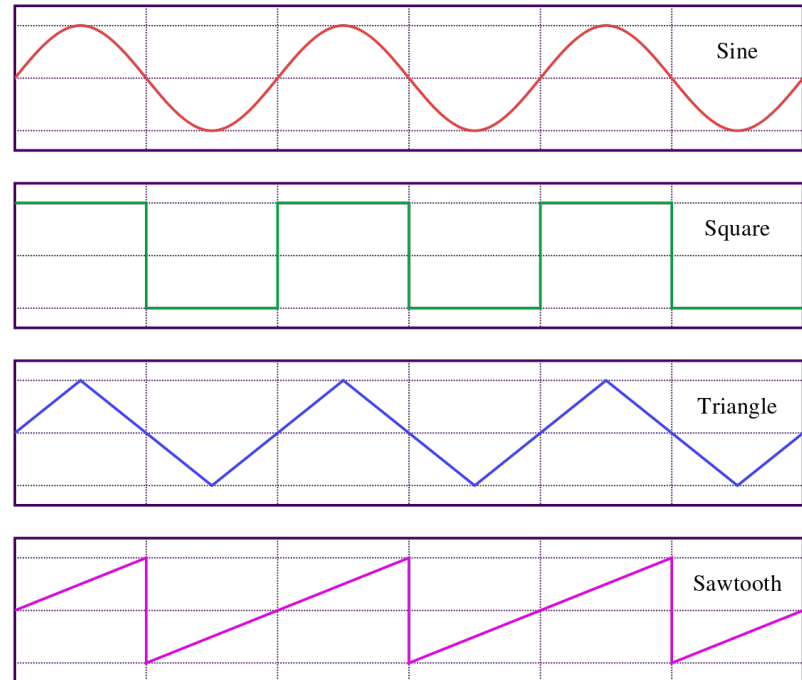
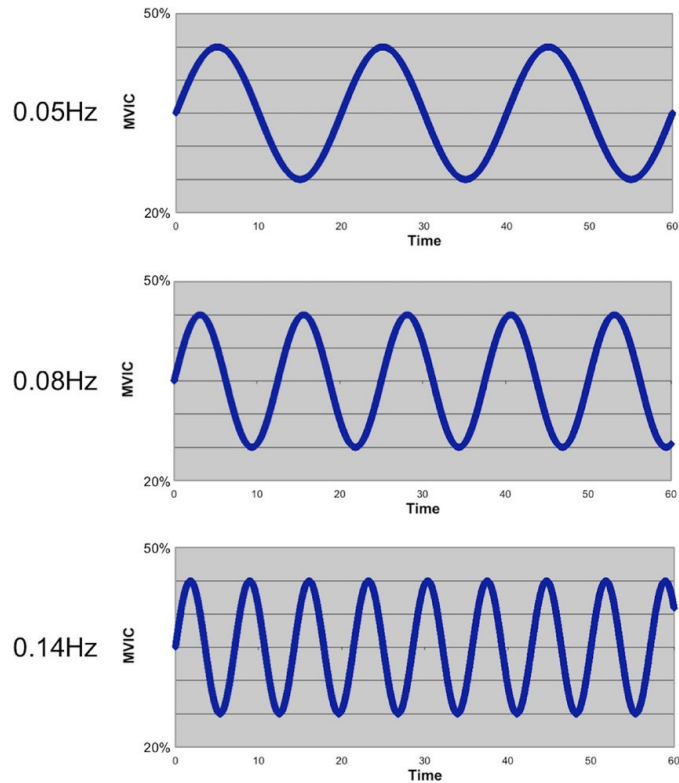


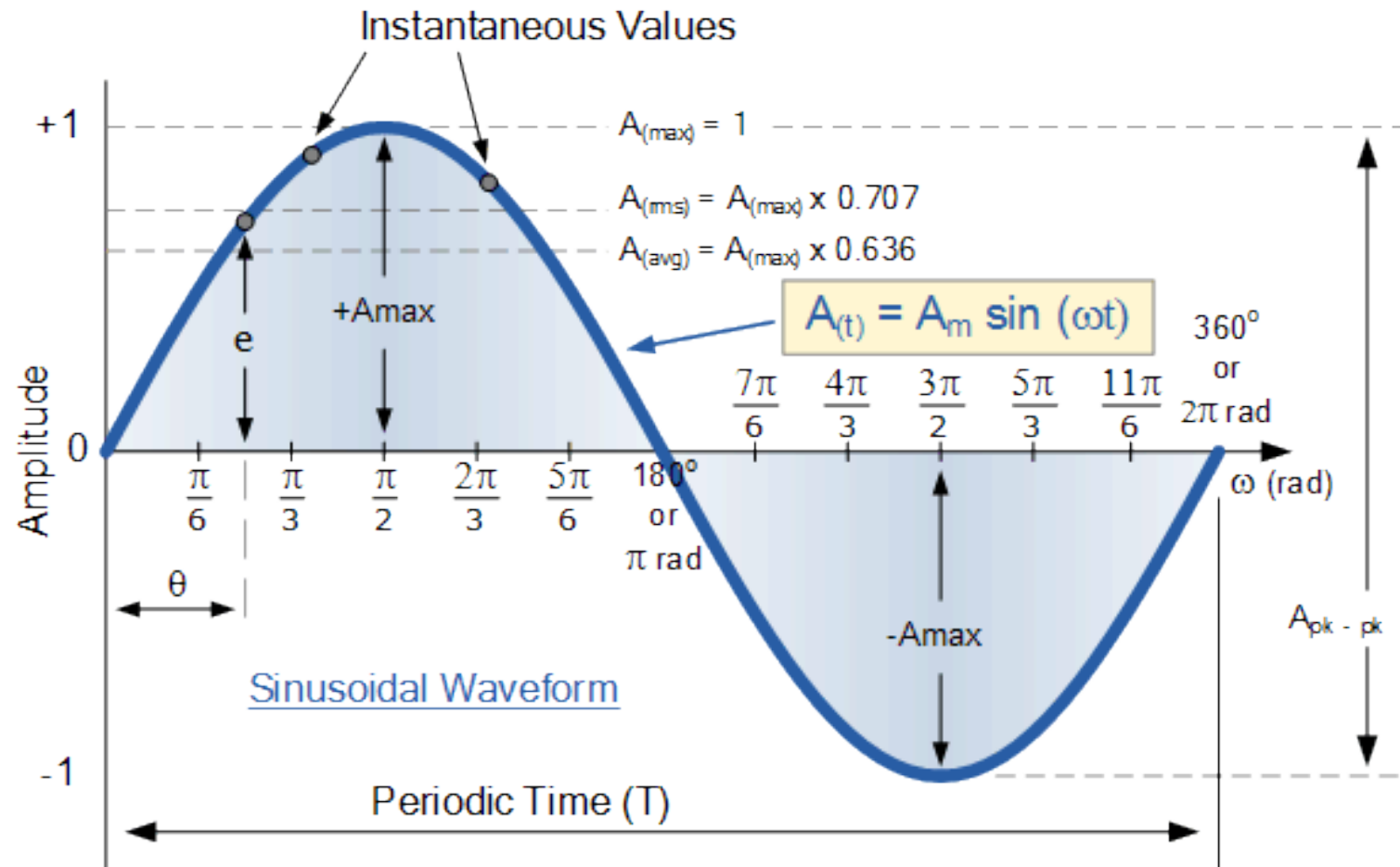
(a) Complete power supply with transformer, rectifier, filter, and regulator

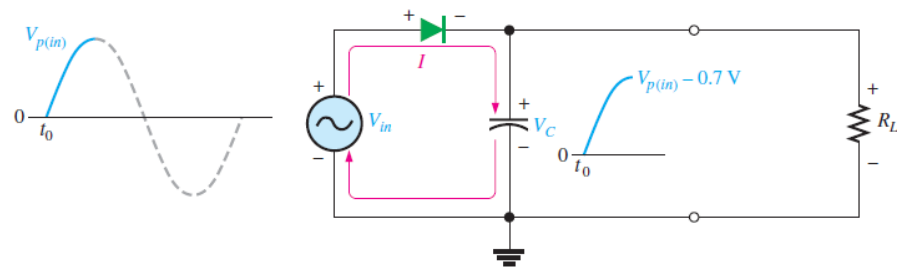


(b) Half-wave rectifier

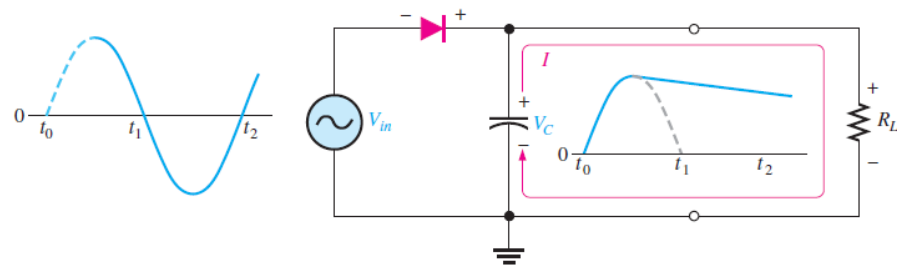
AC Wave Forms



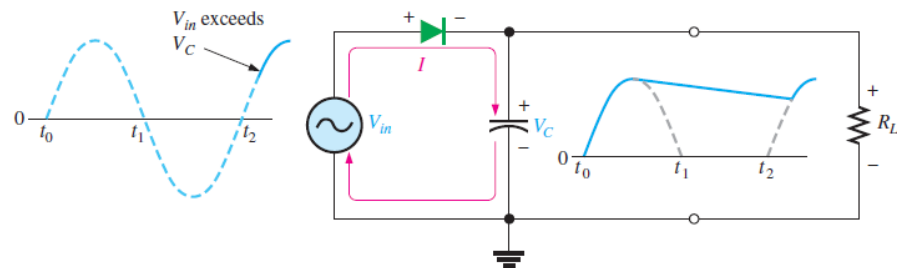


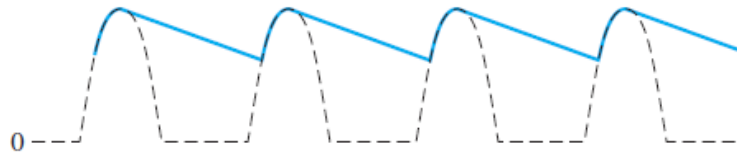


(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on.

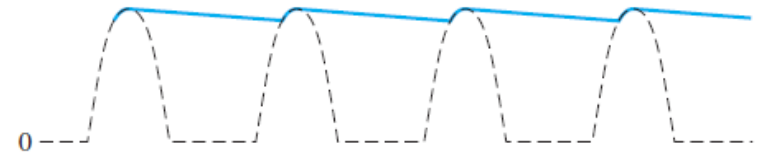


(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.



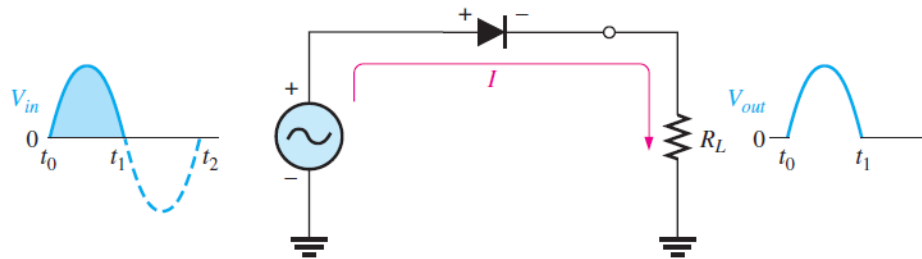


(a) Larger ripple (blue) means less effective filtering.

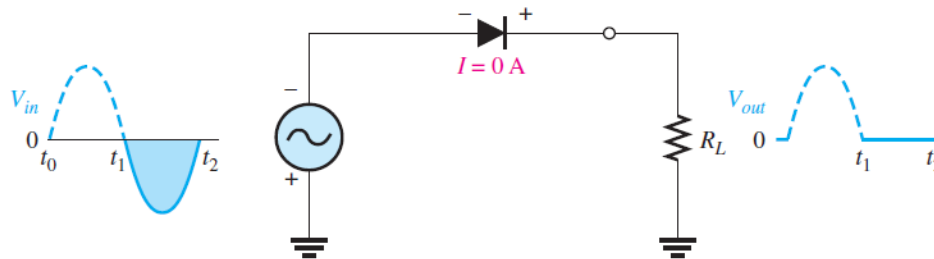


(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

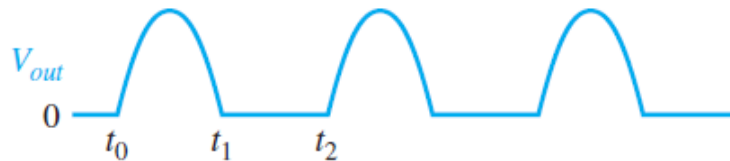
Half-Wave Rectifier



(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



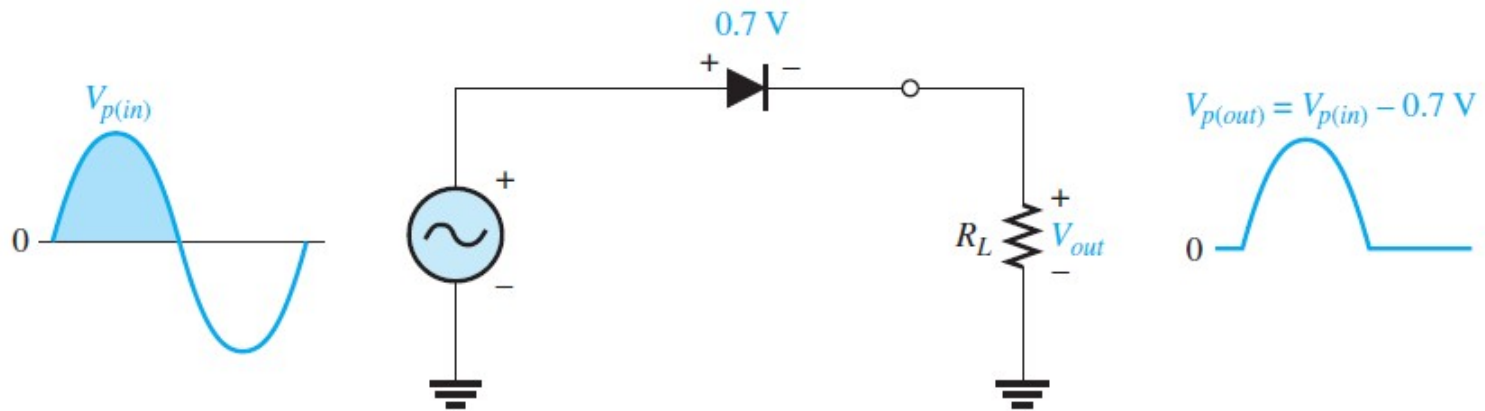
(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.



(c) 60 Hz half-wave output voltage for three input cycles

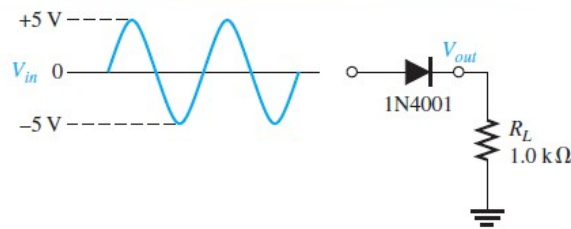
Barrier potential

$$V_{p(out)} = V_{p(in)} - 0.7 \text{ V}$$

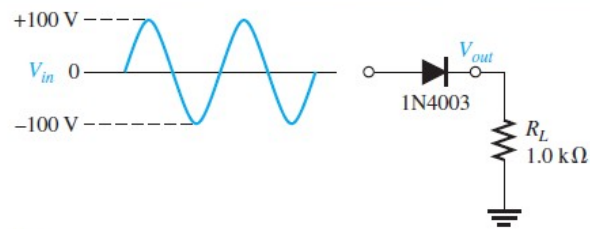


Half-Wave Rectifier

Draw the output voltages of each rectifier for the indicated input voltages, as shown in Figure 2–24. The 1N4001 and 1N4003 are specific rectifier diodes.



(a)



(b)

▲ FIGURE 2–24

Solution The peak output voltage for circuit (a) is

$$V_{p(out)} = V_{p(in)} - 0.7 \text{ V} = 5 \text{ V} - 0.7 \text{ V} = 4.30 \text{ V}$$

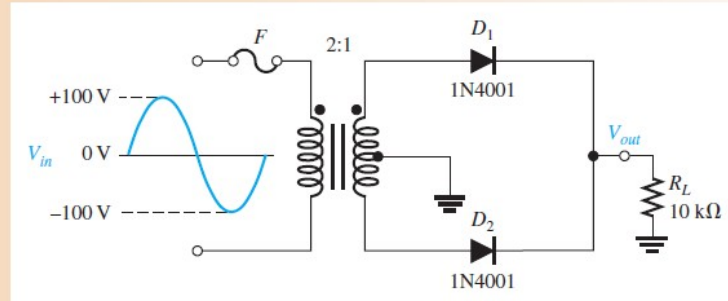
The peak output voltage for circuit (b) is

$$V_{p(out)} = V_{p(in)} - 0.7 \text{ V} = 100 \text{ V} - 0.7 \text{ V} = 99.3 \text{ V}$$

The output voltage waveforms are shown in Figure 2–25. Note that the barrier potential could have been neglected in circuit (b) with very little error (0.7 percent); but, if it is neglected in circuit (a), a significant error results (14 percent).



Center-Tapped Full-Wave Rectifier



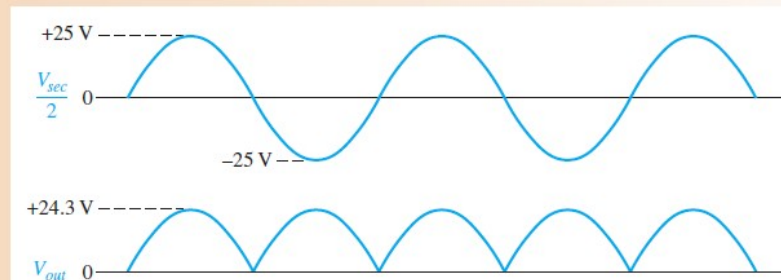
Solution (a) The transformer turns ratio $n = 0.5$. The total peak secondary voltage is

$$V_{p(sec)} = nV_{p(pri)} = 0.5(100 \text{ V}) = 50 \text{ V}$$

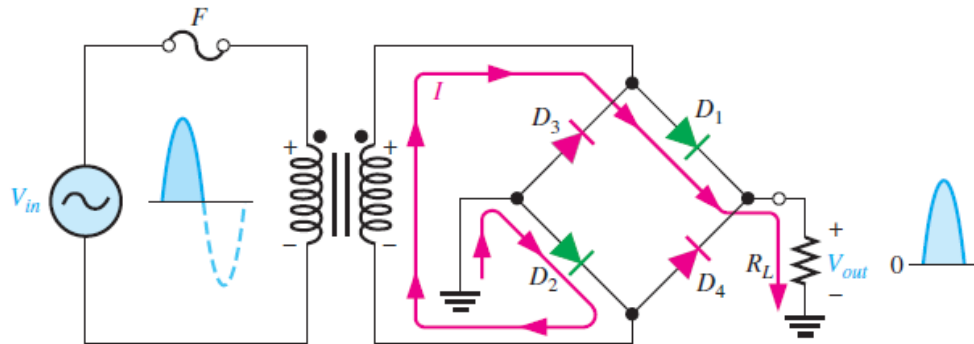
There is a 25 V peak across each half of the secondary with respect to ground. The output load voltage has a peak value of 25 V, less the 0.7 V drop across the diode. The waveforms are shown in Figure 2-37.

(b) Each diode must have a minimum PIV rating of

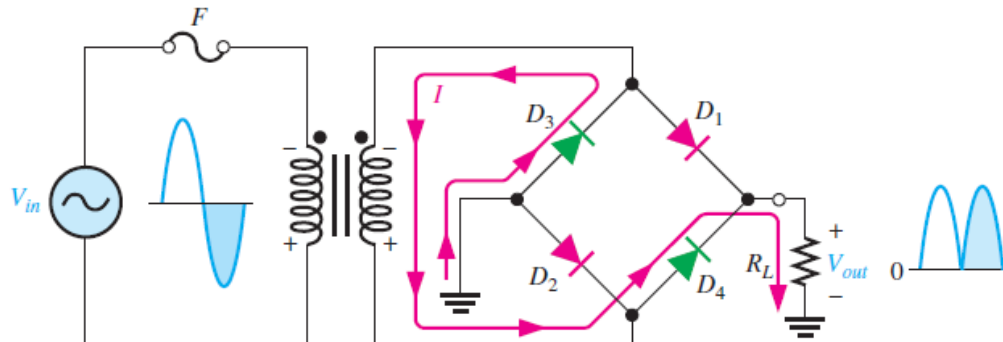
$$\text{PIV} = 2V_{p(out)} + 0.7 \text{ V} = 2(24.3 \text{ V}) + 0.7 \text{ V} = 49.3 \text{ V}$$



Bridge Full-Wave Rectifier

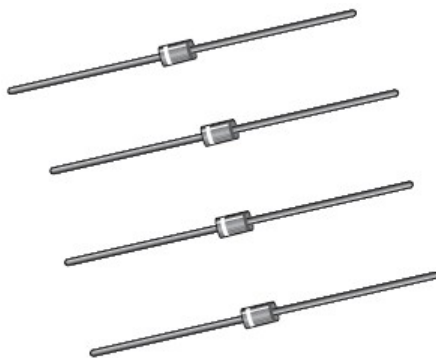
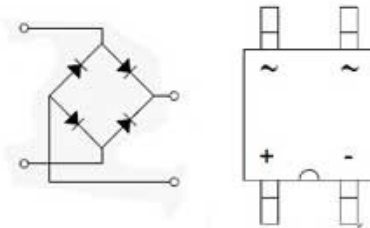
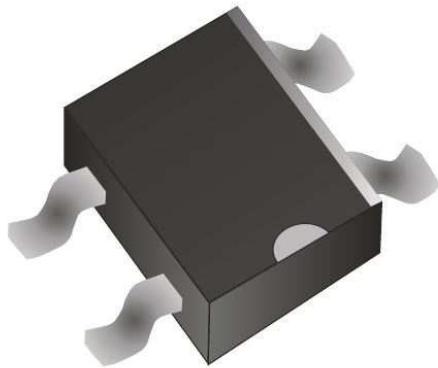


(a) During the positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current. D_3 and D_4 are reverse-biased.



(b) During the negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current. D_1 and D_2 are reverse-biased.

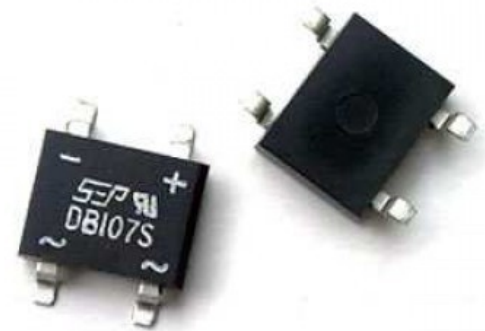
Bridge Rectifier Diode

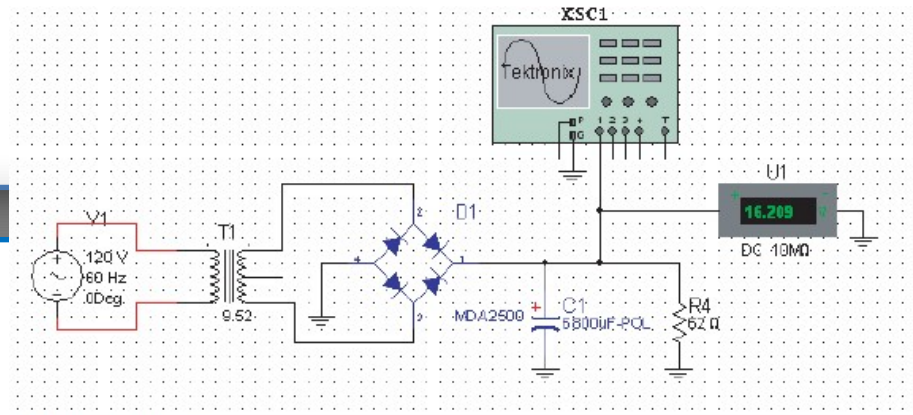


(a) Separate rectifier diodes

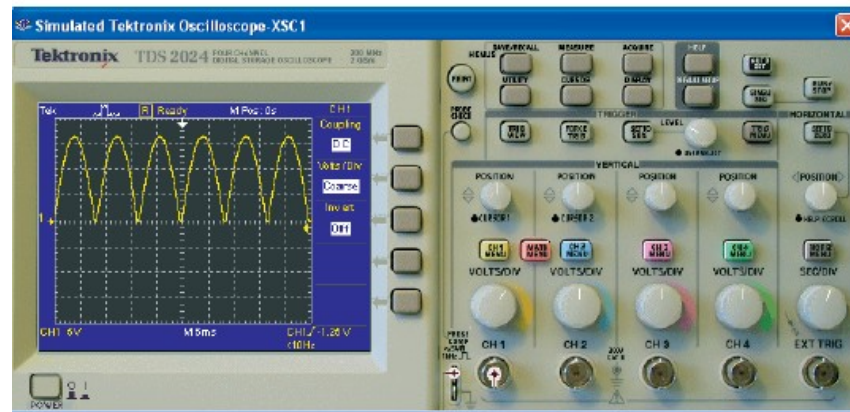


(b) Full-wave bridge rectifier

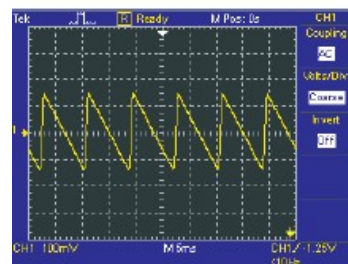




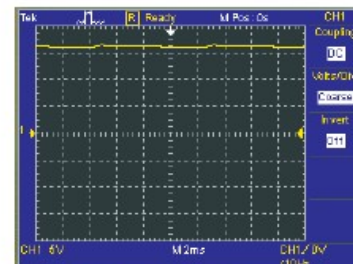
(a) Multisim circuit screen



(b) Output voltage without the filter capacitor

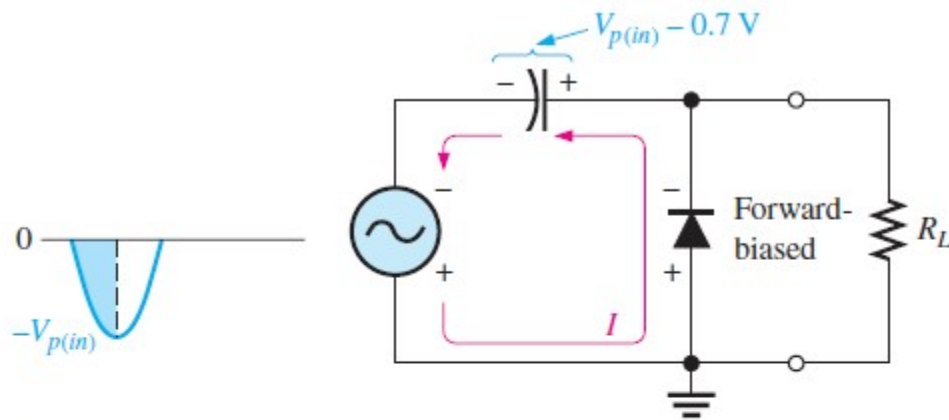


(c) Ripple voltage is less than 300 mV pp

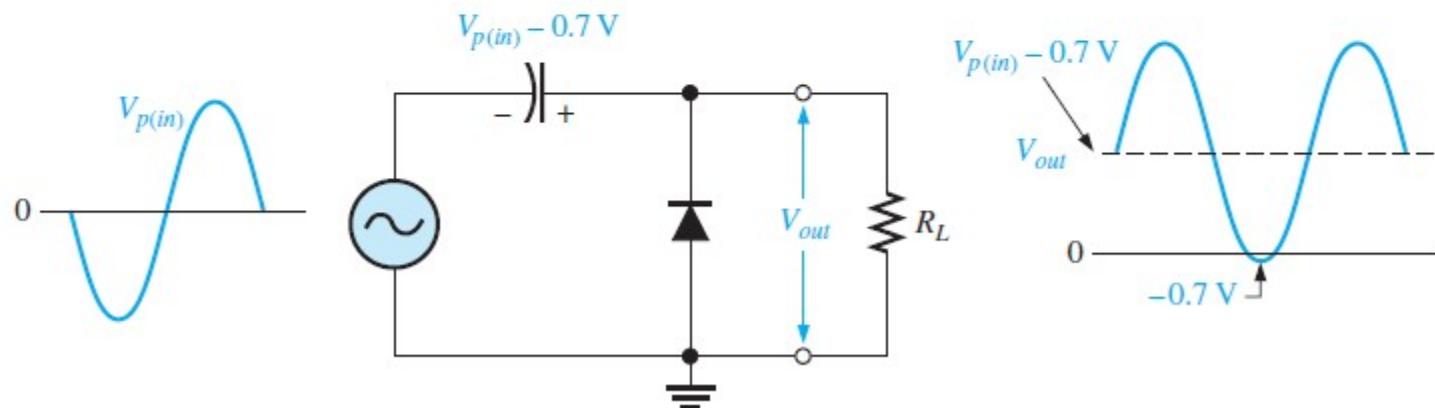


(d) DC output voltage with filter capacitor
(near top of screen)

Diode Clampers (Voltaj Kenetleyiciler)



(a)



(b)

Voltage Multipliers

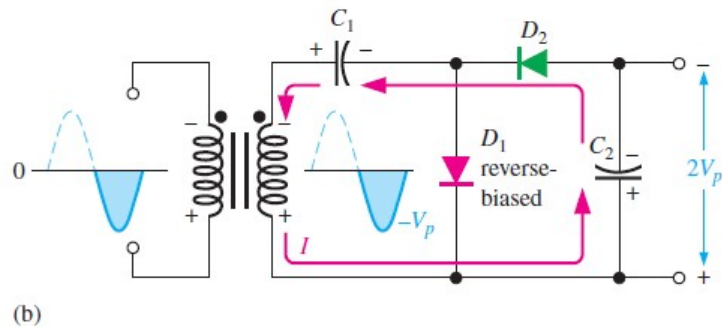
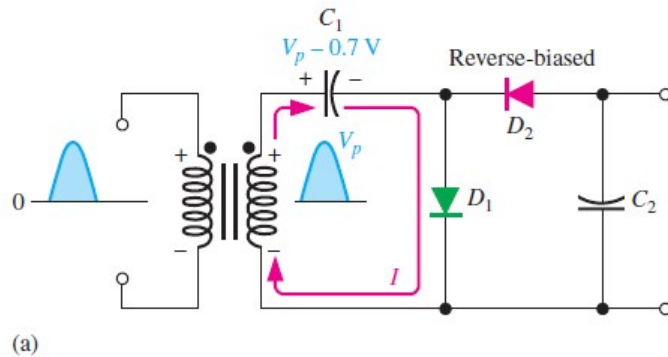
- Cathode-Ray Tubes (CRTs)-high-voltage, low-current
- Discharge Lamps-Gas Lamps

$$V_{C1} - V_{C2} + V_p = 0$$

$$V_{C2} = V_p + V_{C1}$$

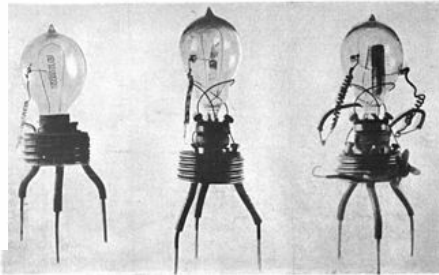
Neglecting the diode drop of D_2 , $V_{C1} = V_p$. Therefore,

$$V_{C2} = V_p + V_p = 2V_p$$



Edison effect, flaming valve, grid audion

1904

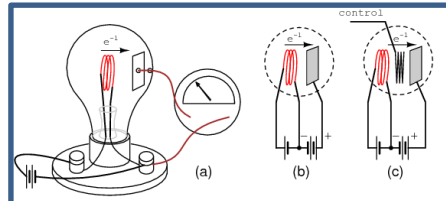


Fleming valve

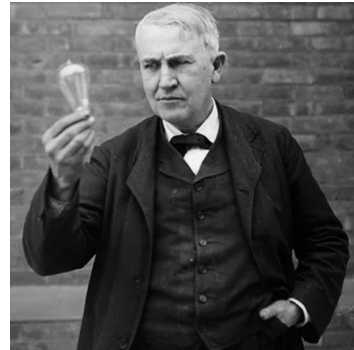


John Ambrose Fleming

Marconi Wireless Telegraph Company



Edison Effect

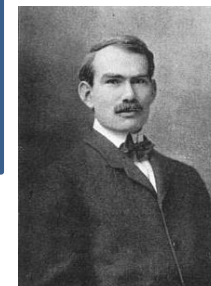


1875

1906



The grid Audion



Lee DeForest

American DeForest Wireless Telegraph Company

Sources: <http://www.allaboutcircuits.com> wikipedia.org

Vacuum Tubes



Diode Datasheet

V_{RRM} The peak reverse voltage that can be applied repetitively across the diode. Notice that it is 50 V for the 1N4001 and 1000 V for the 1N4007. This rating is the same as the PIV.

$I_{F(AV)}$ The maximum average value of a 60 Hz half-wave rectified forward current. This current parameter is 1.0 A for all of the diode types and is specified for an ambient temperature of 75°C.

I_{FSM} The maximum peak value of nonrepetitive single half-sine-wave forward surge current with a duration of 8.3 ms. This current parameter is 30 A for all of the diode types.

T_{stg} The allowable range of temperatures at which the device can be kept when not operating or connected to a circuit.

T_J The allowable range of temperatures for the *pn* junction when the diode is operated in a circuit.

1N4001, 1N4002, 1N4003, 1N4004, 1N4005, 1N4006, 1N4007

MAXIMUM RATINGS

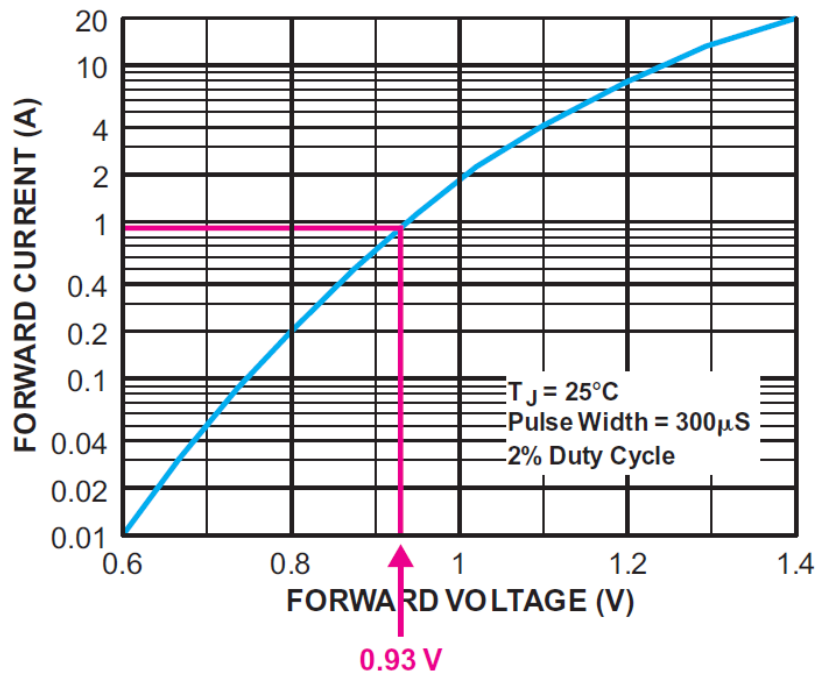
Rating	Symbol	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	Unit
†Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	50	100	200	400	600	800	1000	V
†Non-Repetitive Peak Reverse Voltage (halfwave, single phase, 60 Hz)	V_{RSM}	60	120	240	480	720	1000	1200	V
†RMS Reverse Voltage	$V_{R(RMS)}$	35	70	140	280	420	560	700	V
†Average Rectified Forward Current (single phase, resistive load, 60 Hz, $T_A = 75^\circ\text{C}$)	I_O	1.0							A
†Non-Repetitive Peak Surge Current (surge applied at rated load conditions)	I_{FSM}	30 (for 1 cycle)							A
Operating and Storage Junction Temperature Range	T_J T_{stg}	-65 to +175							$^\circ\text{C}$

Maximum ratings are those values beyond which device damage can occur. Maximum ratings applied to the device are individual stress limit values (not normal operating conditions) and are not valid simultaneously. If these limits are exceeded, device functional operation is not implied, damage may occur and reliability may be affected.

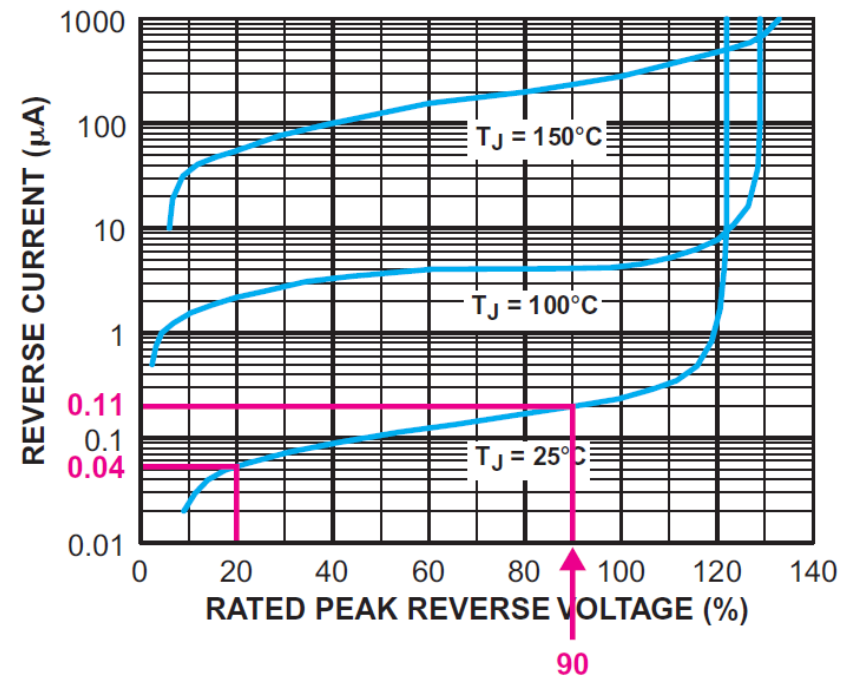
ELECTRICAL CHARACTERISTICS†

Rating	Symbol	Typ	Max	Unit
Maximum Instantaneous Forward Voltage Drop, ($I_F = 1.0$ Amp, $T_J = 25^\circ\text{C}$)	v_F	0.93	1.1	V
Maximum Full-Cycle Average Forward Voltage Drop, ($I_O = 1.0$ Amp, $T_L = 75^\circ\text{C}$, 1 inch leads)	$V_{F(AV)}$	–	0.8	V
Maximum Reverse Current (rated DC voltage) ($T_J = 25^\circ\text{C}$) ($T_J = 100^\circ\text{C}$)	I_R	0.05 1.0	10 50	μA
Maximum Full-Cycle Average Reverse Current, ($I_O = 1.0$ Amp, $T_L = 75^\circ\text{C}$, 1 inch leads)	$I_{R(AV)}$	–	30	μA

Forward Characteristics

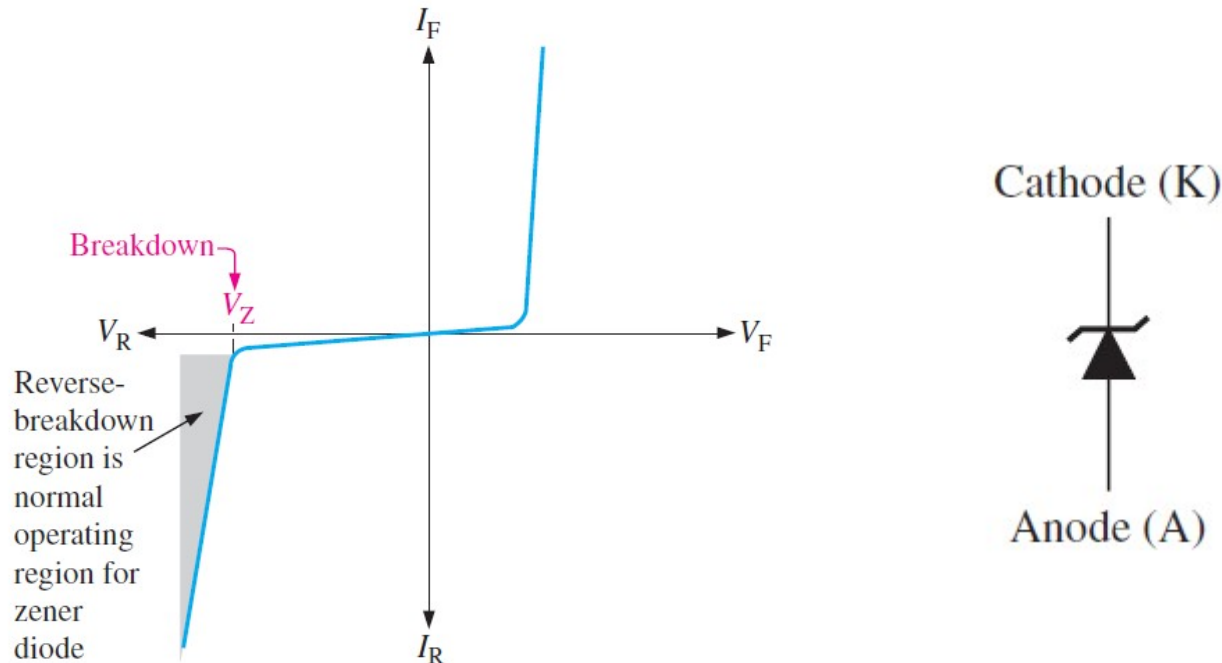


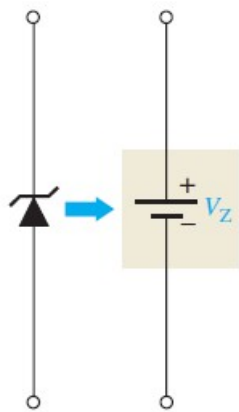
Reverse Characteristics



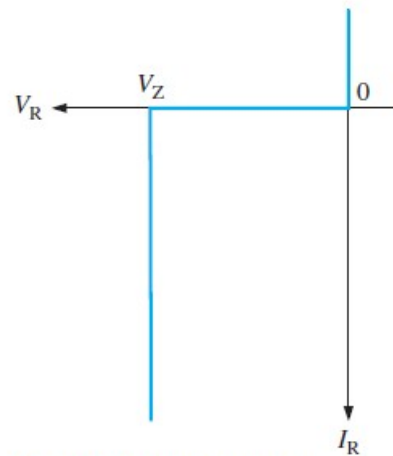
Zener Diode

- Zener diyotlar güç kaynaklarında, voltmetrelerde ve diğer cihazlarda kararlı referans düzgün referans gerilimleri sağlamak için bir gerilim regülatörü olarak kullanılmaktadır

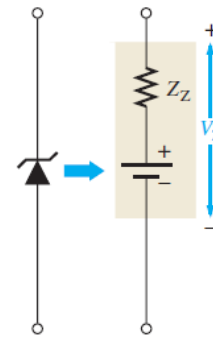




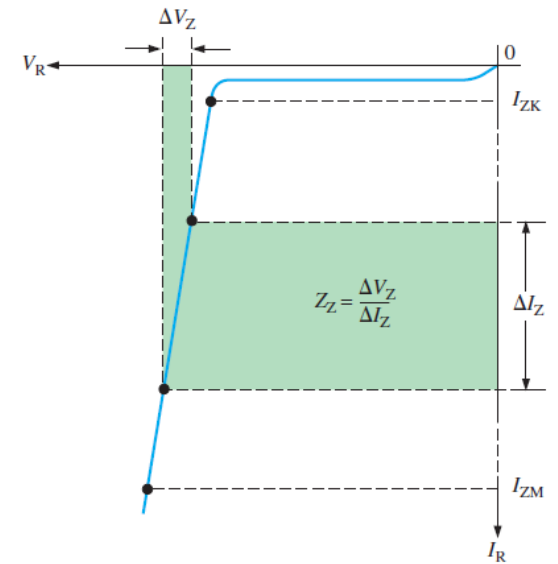
(a) Ideal model



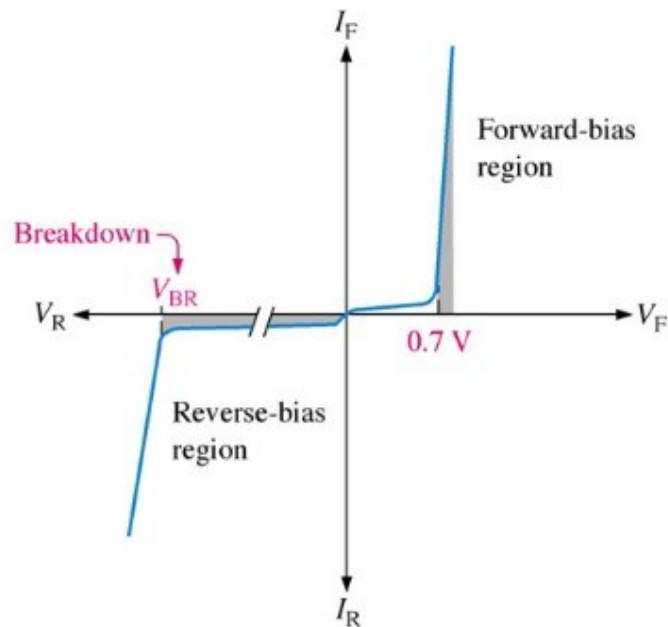
(b) Ideal characteristic curve



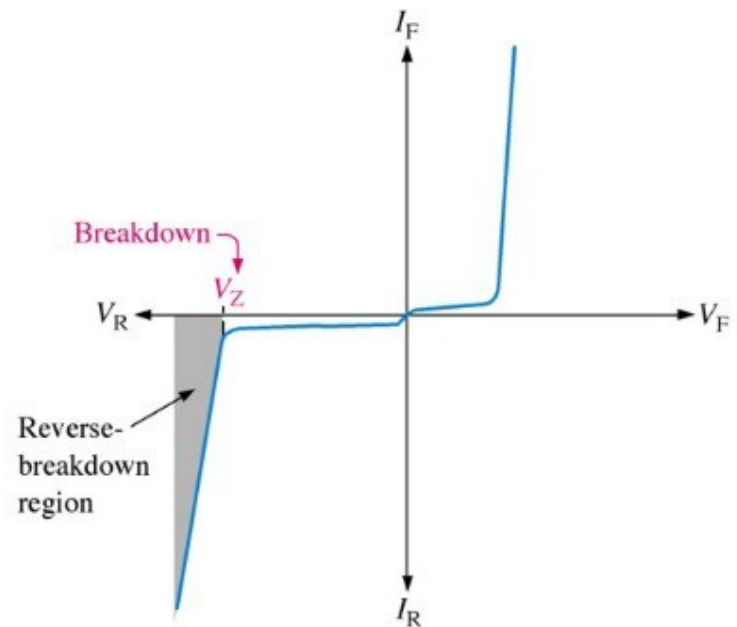
(a) Practical model



(b) Characteristic curve. The slope is exaggerated for illustration.



(a) The normal operating regions for a rectifier diode are shown as shaded areas.



(b) The normal operating region for a zener diode is shaded.

A zener diode is much like a normal diode. The exception being is that it is placed in the circuit in reverse bias and operates in reverse breakdown. This typical characteristic curve illustrates the operating range for a zener. Note that it's forward characteristics are just like a normal diode.

- Normal diyot ileri öngerilimli (polarmalı) olarak çalıştırılırken, Zener diyeti ileri ve geri öngerilimli olarak çalışan özel amaçlı bir diyottur.
- Normal Diyot, akımı sadece bir yönde ileten tek yönlü bir cihaz, Zener diyeti, iki yönlü bir cihazdır.
- Dopingleme açısından, normal bir PN eklem diyeti, Zener diyetuna kıyasla daha az katkı, Zener diyet ise yüksek katkı yarı iletken diyottur.
- Normal bir diyet çok düşük arıza voltajına sahiptir, daha az miktarda ters voltaja izin verir. Zener diyetunun çok yüksek bir arıza voltajı vardır, bu da normal diyetle karşılaştırıldığında daha fazla ters voltaja izin veriri.
- Uygulamaları açısından, doğrultma,kırpma, gerilim çoğaltma işlemlerinde normal bir diyet kullanılırken, gerilim regülatörü olarak Zener diyet kullanılır.

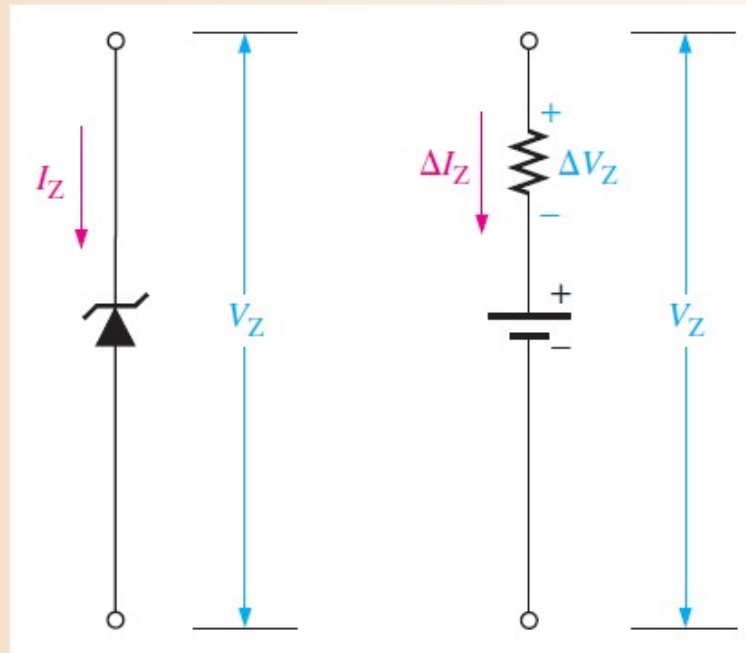
Electrical Characteristics The first column in the datasheet lists the zener type numbers, 1N4728A through 1N4764A.

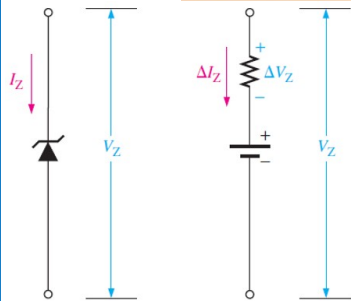
Zener voltage, V_Z , and zener test current, I_Z For each device type, the minimum, typical, and maximum zener voltages are listed. V_Z is measured at the specified zener test current, I_Z . For example, the zener voltage for a 1N4728A can range from 3.315 V to 3.465 V with a typical value of 3.3 V at a test current of 76 mA.

Maximum zener impedance Z_Z is the maximum zener impedance at the specified test current, I_Z . For example, for a 1N4728A, Z_Z is 10 Ω at 76 mA. The maximum zener impedance, Z_{ZK} , at the knee of the characteristic curve is specified at I_{ZK} , which is the current at the knee of the curve. For example, Z_{ZK} is 400 Ω at 1 mA for a 1N4728A.

Leakage current Reverse leakage current is specified for a reverse voltage that is less than the knee voltage. This means that the zener is not in reverse breakdown for these measurements. For example I_R is 100 μA for a reverse voltage of 1 V in a 1N4728A.

From the datasheet in Figure 3–7, a 1N4736A zener diode has a Z_Z of $3.5\ \Omega$. The datasheet gives $V_Z = 6.8\ \text{V}$ at a test current, I_Z , of $37\ \text{mA}$. What is the voltage across the zener terminals when the current is $50\ \text{mA}$? When the current is $25\ \text{mA}$? Figure 3–8 represents the zener diode.





zener test current, I_Z

Zener voltage, V_Z

Solution

For $I_Z = 50$ mA: The 50 mA current is a 13 mA increase above the test current, I_Z , of 37 mA.

$$\Delta I_Z = I_Z - 37 \text{ mA} = 50 \text{ mA} - 37 \text{ mA} = +13 \text{ mA}$$

$$\Delta V_Z = \Delta I_Z Z_Z = (13 \text{ mA})(3.5 \Omega) = +45.5 \text{ mV}$$

The change in voltage due to the increase in current above the I_Z value causes the zener terminal voltage to increase. The zener voltage for $I_Z = 50$ mA is

$$V_Z = 6.8 \text{ V} + \Delta V_Z = 6.8 \text{ V} + 45.5 \text{ mV} = \mathbf{6.85 \text{ V}}$$

For $I_Z = 25$ mA: The 25 mA current is a 12 mA decrease below the test current, I_Z , of 37 mA.

$$\Delta I_Z = -12 \text{ mA}$$

$$\Delta V_Z = \Delta I_Z Z_Z = (-12 \text{ mA})(3.5 \Omega) = -42 \text{ mV}$$

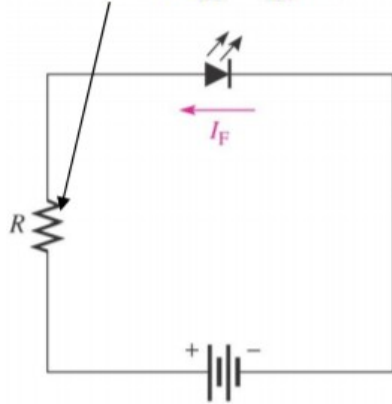
The change in voltage due to the decrease in current below the test current causes the zener terminal voltage to decrease. The zener voltage for $I_Z = 25$ mA is

$$V_Z = 6.8 \text{ V} - \Delta V_Z = 6.8 \text{ V} - 42 \text{ mV} = \mathbf{6.76 \text{ V}}$$

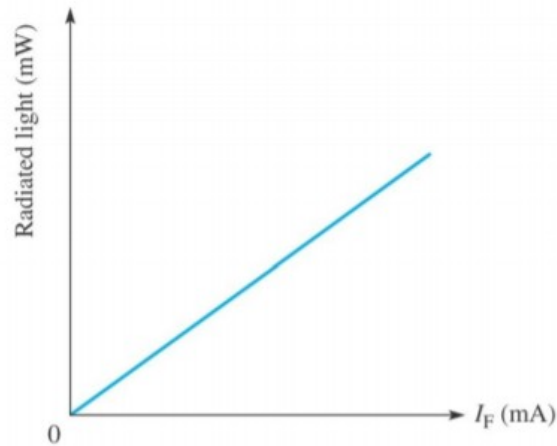
Light-emitting diodes (LEDs)

Always have a
current-limiting
resistor in a LED
circuit.
Usually 1K – 2K
Ohms.

$$R = (V_S - V_L) / I$$



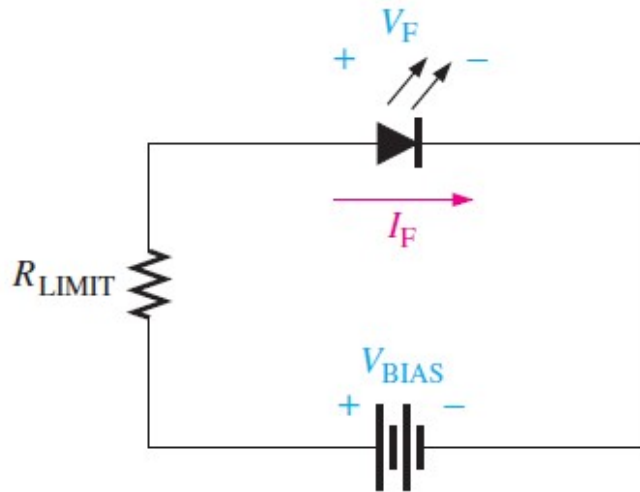
(a) Forward-biased operation



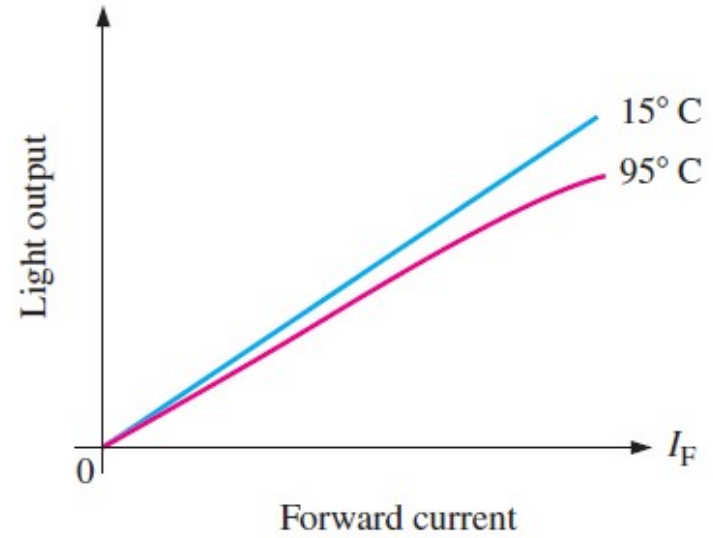
(b) Typical light output versus forward current



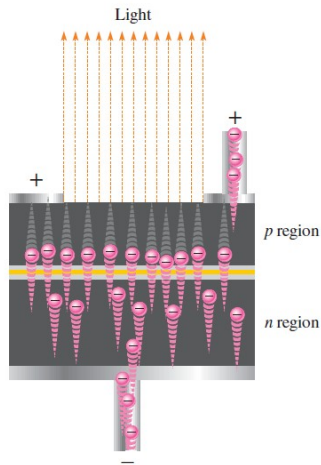
(c) Typical LEDs

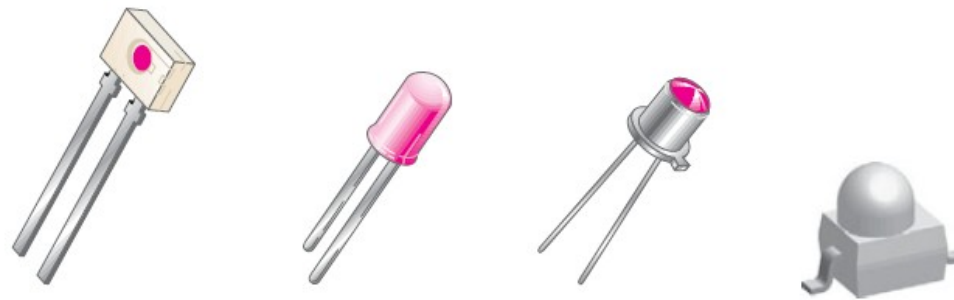


(a) Forward-biased operation



General light output versus forward current for two temperatures

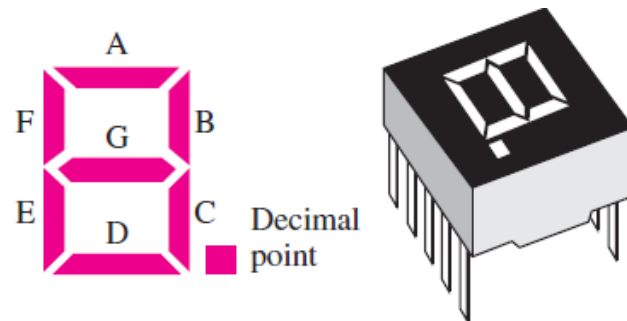




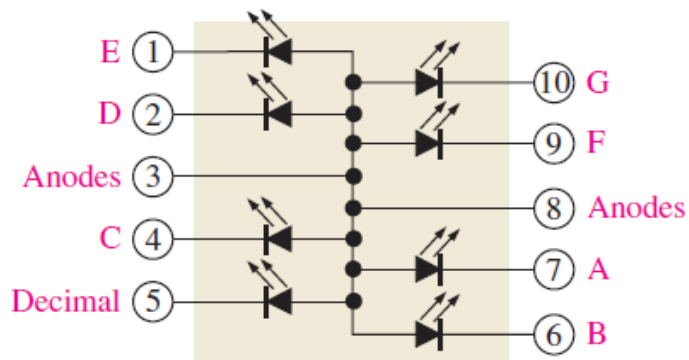
(a) Typical small LEDs for indicators



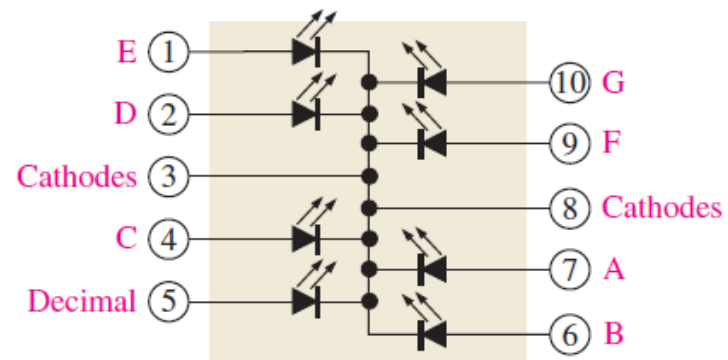
The 7-segment LED display



(a) LED segment arrangement and typical device

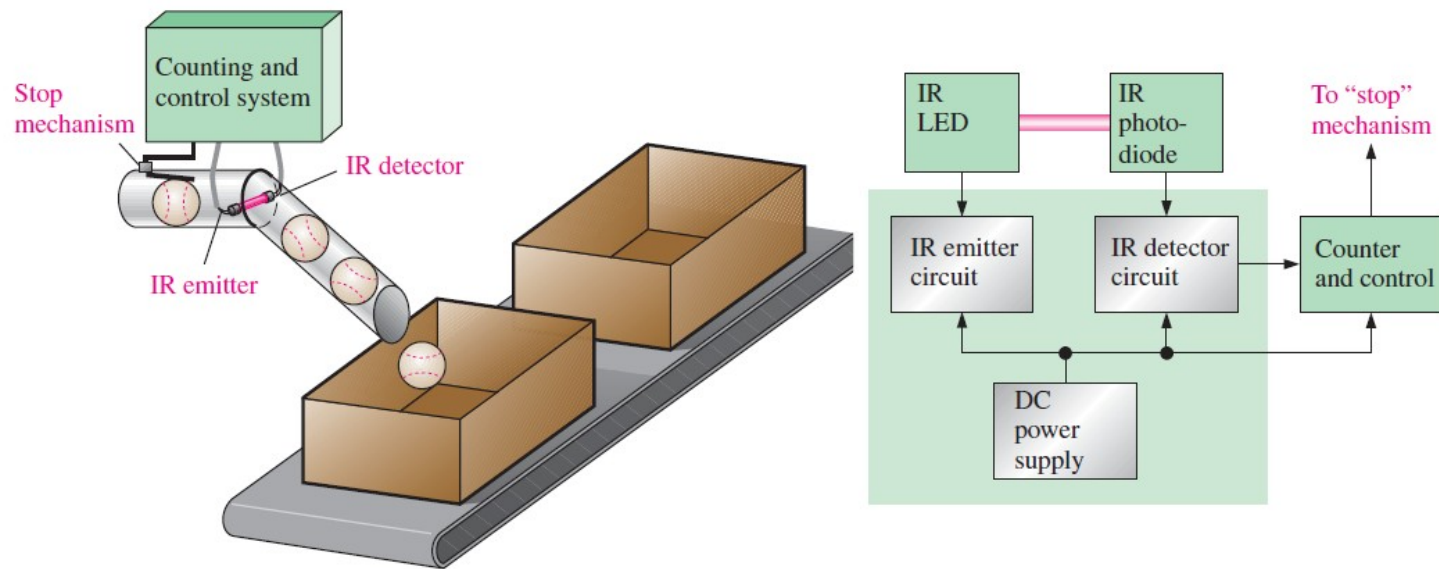


(b) Common anode

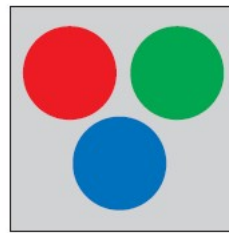


(c) Common cathode

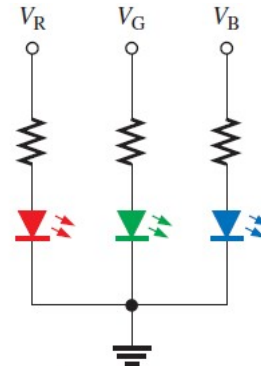
Counter- Control



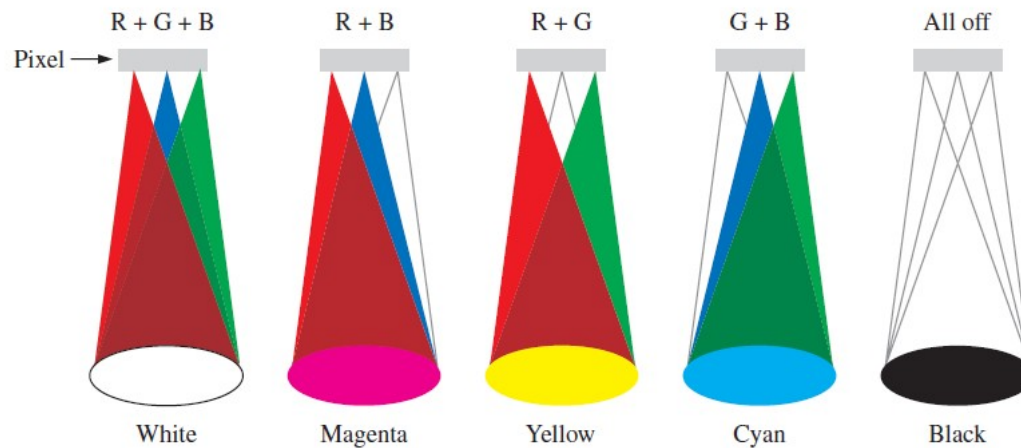
R-G-B



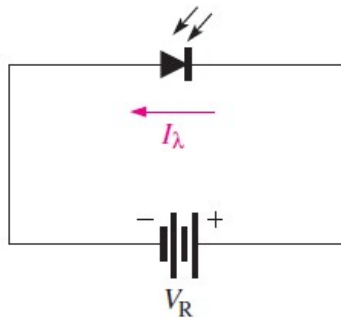
(a) Basic pixel



(b) Pixel circuit



The Photodiode



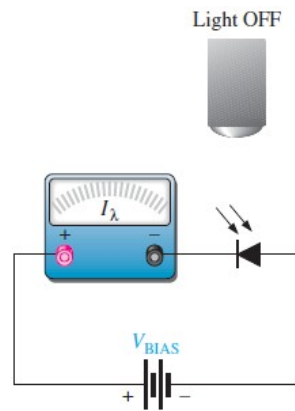
(a) Reverse-bias operation using standard symbol



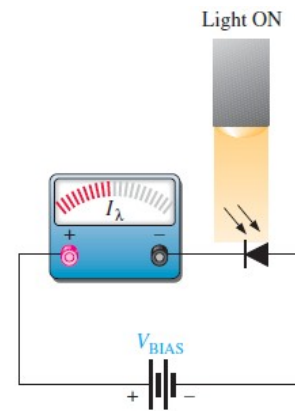
(b) Typical devices



(c) Alternate symbol



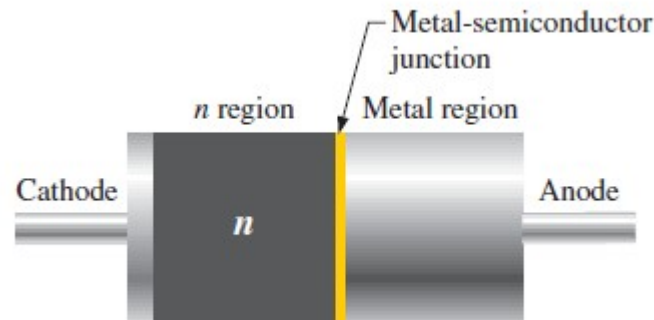
(a) No light, no current except negligible dark current



(b) Where there is incident light, resistance decreases and there is reverse current.

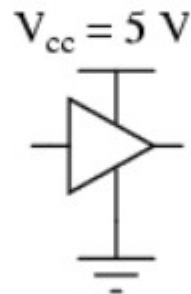
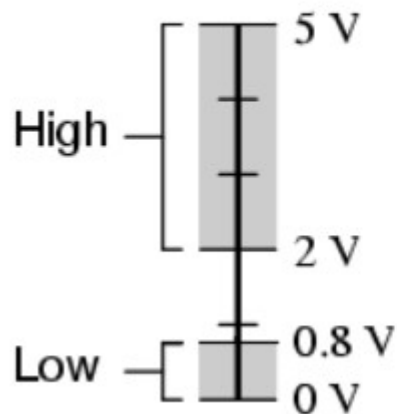
The Schottky Diode

- Schottky diyot, katkılı (doped) yarı iletken bir bölgenin (genellikle n tipi) altın, gümüş veya platin gibi bir metal ile birleştirilmesiyle oluşturulur.
- Schottky diyotunun ters sızıntı akımı yoktur.
- Cevap süresi çok hızlıdır. Anahtarlama sürelerini kısaltmak için yüksek frekanslı uygulamalarda ve birçok dijital devrede kullanılabilir. LS düşük güçlü Schottky anlamına gelir) Schottky diyotunu kullanan bir tür dijital entegre devredir.

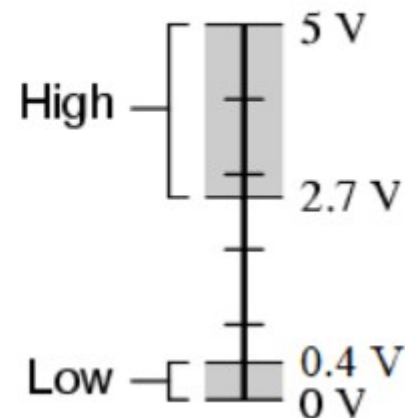


TTL (Transistor-Transistor-Logic)

Acceptable TTL gate input signal levels



Acceptable TTL gate output signal levels



<https://learn.sparkfun.com/tutorials/logic-levels/ttl-logic-levels>

Low-Power Schottky TTL (74LS)

- 74LS00

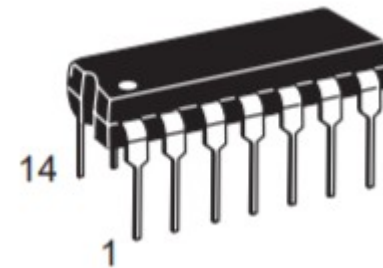
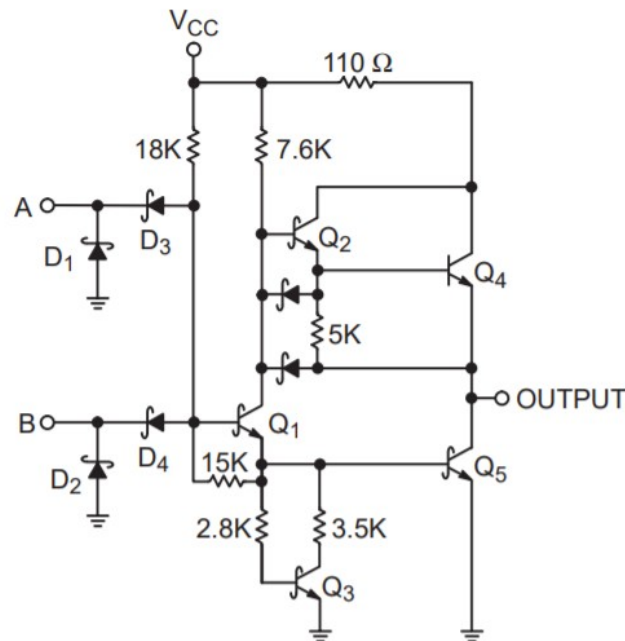
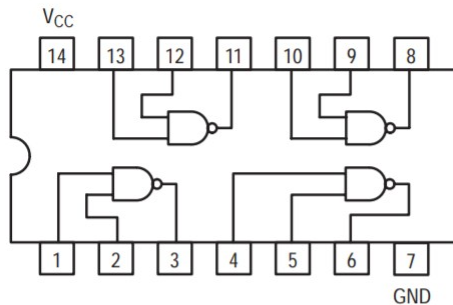
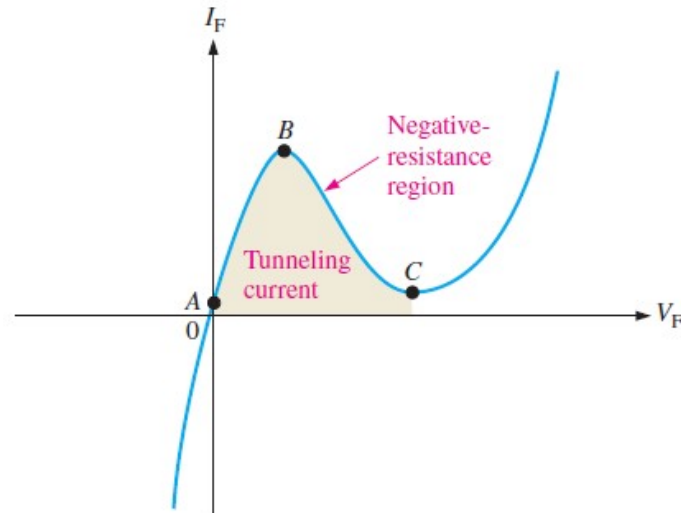
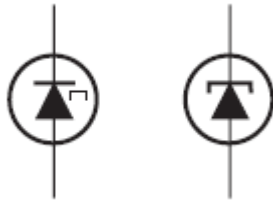


Figure 1. LS00 — 2-Input NAND Gate

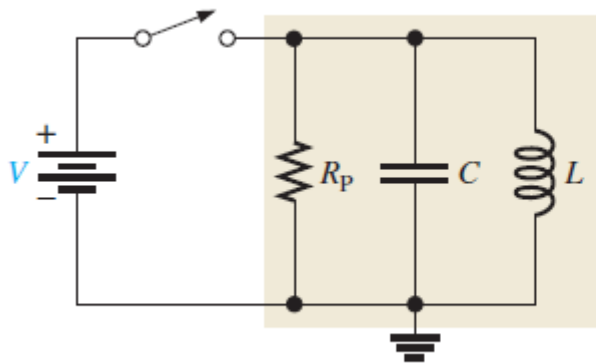
http://ecee.colorado.edu/~mcclurel/ON_Semiconductor_LSTTL_Data_DL121-D.pdf

The Tunnel Diode

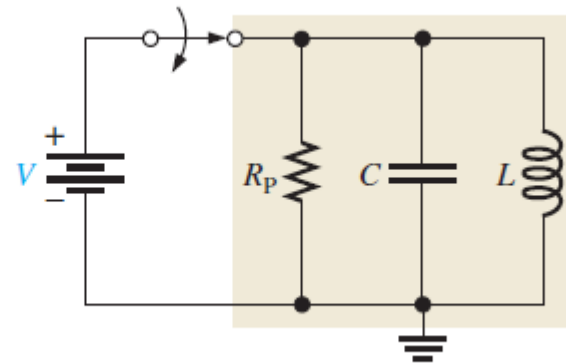
- Tünel diyot, negatif direnç olarak bilinen özel bir özellik sergiler. Bu özellik, osilatör ve mikrodalga amplifikatör uygulamalarında kullanışlı olmasını sağlar. Tünel diyotları, p ve n bölgelerini geleneksel bir diyottan çok daha fazla germanyum veya galyum arsenit katmak suretiyle yapılır. Bu durum, diyotun tüm ters gerilimler için iletme izin vermesi anlamına gelir. Böylece geleneksel doğrultucu diyotta olduğu gibi herhangi bir arıza etkisi olmaz.



Parallel resonant circuit



(a)



(b)

